

Florian Julé

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Work Experience

Joby Aviation

Santa Cruz, CA

Staff Mechanical Engineer – Special Projects, Flight Research

Nov 2023 – Feb 2025

- Designed and tested mechanical systems across novel aircraft programs within Joby's flight research division, including retractable landing gear, structural components, and system integration.

Tail and Effectors IPT Lead Engineer – S4-2.1 production aircraft

Oct 2018 – Nov 2023

- Led cross-functional teams in the development of tail, tilt nacelles, control surfaces, and propellers, advancing airframe designs towards FAA certification and production. 5 direct reports.
- Delivered the first production-prototype aircraft and conforming drawing package.
- Granted patent for the inboard nacelle tilt mechanism: [Rotor assembly deployment mechanism](#).

Tilt Nacelles Lead Engineer – S4-2.0, pre-production aircraft

Oct 2017 – Oct 2018

- Designed, built, and tested the tilt nacelles for the Joby S4 pre-production aircraft.
- Served as flight test engineer throughout the flight test program.

Mechanical Engineer – S4-1.0, first full-scale eVTOL aircraft

Feb 2017 – Oct 2017

- Supported composites manufacturing and structural testing for the first Joby S4 full-scale prototype.

Projects

Driver-Assisted Stair-Climbing Wheelchair

Apr 2026 – Present

- Designing and building a 4WD articulated chassis with a passive yaw joint and a self-leveling seat for stair-climbing and outdoor navigation; ROS 2 stack (C++/Python) on a Jetson Orin Nano with a Gazebo digital twin.
- The Advanced driver-assistance system uses two cameras. One Intel RealSense D435i that runs with YOLO26 semantic segmentation trained on ADE20K dataset for general navigation including obstacle and stair detection. A second D405 camera uses depth to handle stair navigation.

Cooperative Payload Transport with a Drone Swarm

Jan – Mar 2026

- Formulated and solved an Optimal Control Problem (OCP) with CasADi to generate collision-free, dynamically feasible trajectories; deployed and validated on real hardware via ROS 2 and Crazyswarm2.

EKF SLAM on a TurtleBot3

Jan – Mar 2026

- Built a full SLAM stack from scratch in C++/ROS 2: 2D geometry library, simulator, odometry, and Extended Kalman Filter SLAM with unknown data association using LIDAR landmark detection.

Object Manipulation with Franka Arm

Nov – Dec 2025

- Developed pick-and-place software for a Franka robotic arm using YOLO-based object detection; led computer vision and system integration on real hardware in a team of 4.

Education

Northwestern University

M.S. in Robotics

Evanston, IL

Sep 2025 – Sep 2026

University of Michigan

M.S. in Aerospace Engineering

Ann Arbor, MI

Sep 2015 – Dec 2016

Arts & Métiers ParisTech

B.S. in Mechanical & Industrial Engineering

Metz/Paris, France

Sep 2013 – Jun 2015

Skills

Robotics	Localization (EKF, UKF, Kalman Filters), SLAM, Motion Planning (RRT, A*), Computer Vision, Controls (PID, Feedforward)
Software	Python, C, C++, ROS 2, Git, Matlab
Mechanical	CAD (3D Experience, Onshape), FEA (Abaqus, Simulia), GD&T (ASME Y14.5)
Manufacturing	Composites (hand lay-up, AFP), Metals (CNC, 3D printing, sheet metal), Prototyping
Languages	French, English